



Preliminary Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- In Questions 8–13, show relevant mathematical reasoning and/or calculations

Total Marks – 85

Section I

Pages 1–3

7 marks

- Attempt Questions 1–7
- Allow about 10 minutes for this section

Section II

Pages 4–9

78 marks

- Attempt Questions 8–13
- Allow about 1 hour and 50 minutes for this section

Student Name: _____

Student Number: _____

Teacher:

- ☐ Mr Trenwith
☐ Ms Lee
☐ Mr Moon
☐ Mrs Juhn

- ☐ Ms Everingham
☐ Ms Narayanan
☐ Mrs Kennedy
☐ Mrs Thill

QUESTION	MARK
1–7	/7
8	/13
9	/13
10	/13
11	/13
12	/13
13	/13
TOTAL	/85

Section I

7 marks

Attempt Questions 1–7

Allow about 10 minutes for this section

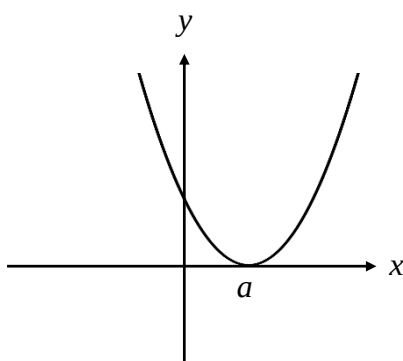
Use the multiple-choice answer sheet for Questions 1–7.

1 What is the value of $\sqrt{\frac{5\pi}{7}}$, correct to 2 significant figures?

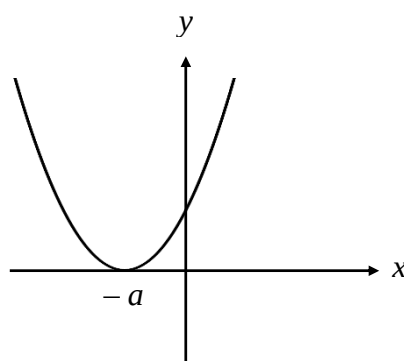
- (A) 1.5
- (B) 1.4
- (C) 1.49
- (D) 1.50

2 Given that $a > 0$, which graph best represents $y = (x - a)^2$?

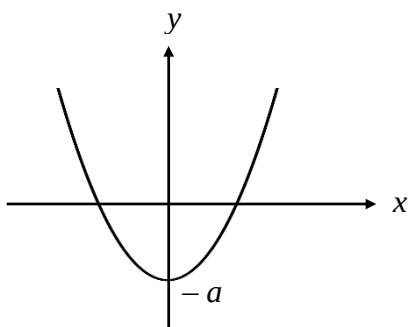
(A)



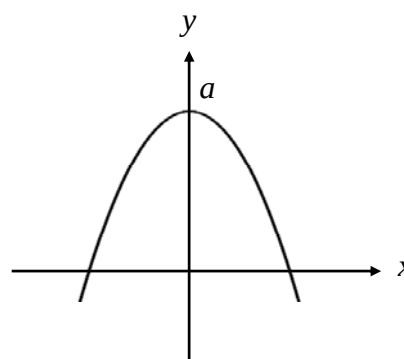
(B)



(C)



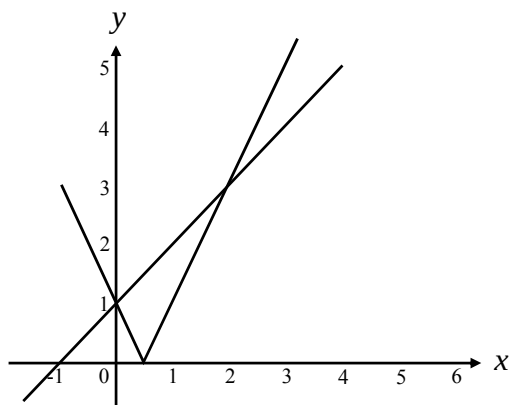
(D)



3 Which of the following is sufficient information to prove that a quadrilateral is a square?

- (A) Opposite sides are parallel.
- (B) Diagonals are perpendicular and opposite sides are parallel.
- (C) Diagonals bisect each other at right angles.
- (D) All vertex angles are 90° and diagonals are perpendicular.

- 4 The graphs of $y = |1 - 2x|$ and $y = x + 1$ are shown below.



What are the solutions to $x + 1 - |1 - 2x| < 0$?

- (A) $0 < x < 2$
 (B) $x < 0$ or $x > 2$
 (C) $1 < x < 3$
 (D) $x < 1$ or $x > 3$
- 5 Which of the following is the derivative of $\sqrt{7x}$?

(A) $\frac{1}{2\sqrt{7x}}$

(B) $\sqrt{7}x$

(C) $\frac{\sqrt{7x}}{2}$

(D) $\frac{1}{2}\sqrt{\frac{7}{x}}$

- 6 The parabola $y = x^2$ is shifted left by 2 units, shifted up by 1 unit and then reflected about the y axis. What is the equation of the parabola now?

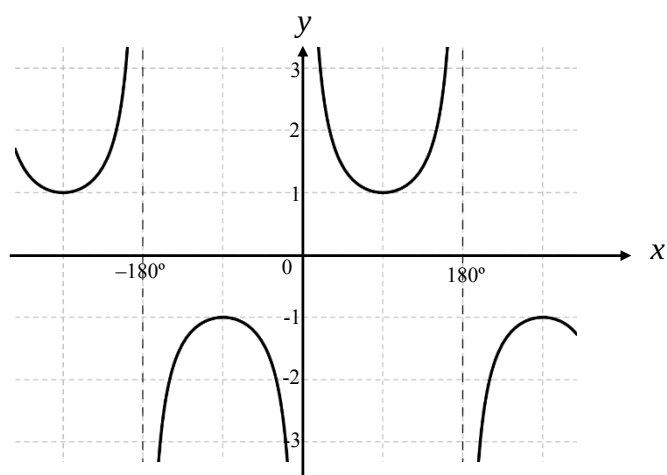
(A) $(x - 2)^2 = y - 1$

(B) $(x + 2)^2 = y + 1$

(C) $(x - 2)^2 = y + 1$

(D) $(x + 2)^2 = y - 1$

7 Which of the following equations best describes the graph below?



(A) $y = \operatorname{cosec} x$

(B) $y = \sec x$

(C) $y = \cot x$

(D) $y = -\sec x$

Section II

78 marks

Attempt Questions 8–13

Allow about 1 hour and 50 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing papers are available.

In Questions 8–13, your responses should include relevant mathematical reasoning and/or calculations.

Question 8 (13 marks) Use a SEPARATE writing booklet.

(a) Factorise $8m^3 - 1$. **2**

(b) Rationalise the denominator of $\frac{\sqrt{3} - \sqrt{2}}{\sqrt{3} + \sqrt{2}}$. **2**

(c) Find the points of intersection of the line $x - y + 3 = 0$ and the hyperbola $xy = 10$. **3**

(d) Find the centre and radius of the circle $x^2 - 6x + y^2 + 4y + 8 = 0$. **3**

(e) For what values of m does the equation $6x^2 - 12mx - 7m + 3 = 0$ have equal roots? **3**

End of Question 8

Question 9 (13 marks) Use a SEPARATE writing booklet.

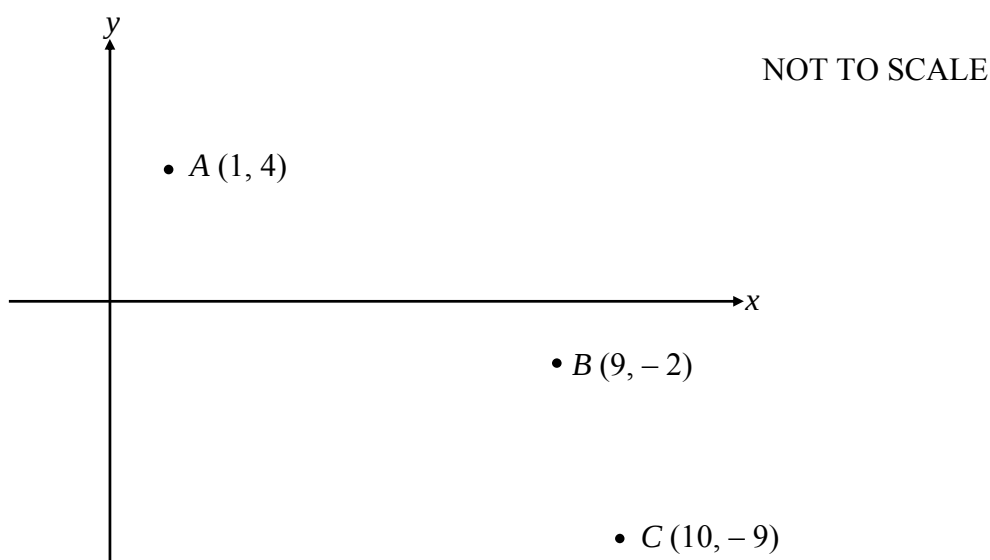
(a) Differentiate:

(i) $y = 2x^3 + 8x$ 1

(ii) $y = \frac{x^2}{1-x}$ 2

(iii) $y = 10x(3x-1)^5$ 3

(b) It is given that points A , B and C have coordinates $(1, 4)$, $(9, -2)$ and $(10, -9)$ respectively.



(i) Find the distance between A and B . 1

(ii) Find the equation of the line that is parallel to AB and passes through C . 2

(iii) Hence, show that the perpendicular distance between C and the line AB is 5 units. 1

(iv) Point D is placed so that $AB \parallel DC$. Given that quadrilateral $ABCD$ has area of 50 u^2 , show that the length of CD is 10 units. 2

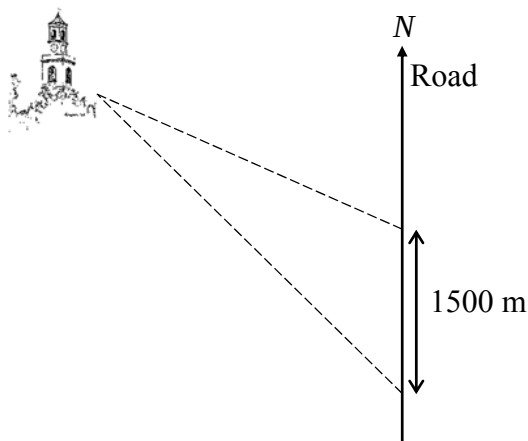
(v) Find the coordinates of D . 1

End of Question 9

Question 10 (13 marks) Use a SEPARATE writing booklet.

- (a) The gradient of the tangent to the curve $y = ax^3 + bx^2$ at $(1, 4)$ is 5. **3**
Find the values of a and b .
- (b) Find all solutions of $\sin 2x = \frac{1}{2}$, where $0^\circ \leq x \leq 360^\circ$. **3**
- (c) Find values of a , b and c , for which $-x^2 + 4x - 3 \equiv a(x+3)^2 + b(x+3) + c$. **2**
- (d) Solve for x : $2^x < 5$. **2**
- (e) A man walking due north along a level road observes a church spire on a bearing of 342° T. After walking 1500 metres, he observes it on a bearing of 337° T. **3**

Copy or trace the diagram into your writing booklet.



Find the perpendicular distance from the church to the road.

End of Question 10

Question 11 (13 marks) Use a SEPARATE writing booklet.

(a) If α and β are the roots of the equation $2x^2 - 3x + 5 = 0$, find the value of the following:

(i) $\alpha + \beta$ **1**

(ii) $\alpha\beta$ **1**

(iii) $(\alpha + 2)(\beta + 2)$ **2**

(b) Prove that $(\sec \theta - \cos \theta)^2 = \tan^2 \theta - \sin^2 \theta$. **3**

(c) (i) Derive the equation of the locus of all points $P(x, y)$ which are equidistant from the origin O and the line $y = 4$. **2**

(ii) Describe the locus geometrically. **1**

(d) Find the values of k for which the expression $kx^2 + (k-1)x + k$ is positive definite. **3**

End of Question 11

Question 12 (13 marks) Use a SEPARATE writing booklet.

(a) Evaluate $\lim_{x \rightarrow 5} \frac{6x^2 - 29x - 5}{x - 5}$ 2

(b) Sketch $y = |x^2 - 5| + 1$, showing all important features. 2

(c) Consider the function $f(x) = x^2 - 5x + 2$.

(i) Differentiate $f(x)$ from first principles. 2

(ii) Find the equation of the normal to the curve $y = f(x)$ at the point where $x = 1$. 2

(d) A parabola whose axis is parallel to the x -axis has vertex $(2, 3)$ and passes through $(-1, 9)$. 2
Find the equation of the parabola.

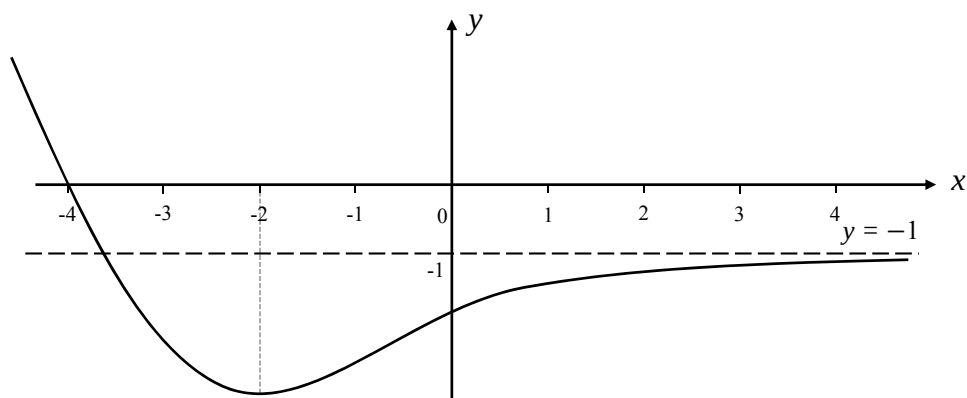
(e) Two cars leave a point P at the same time. They travel away from each other on different straight roads. One car travels at 10 km/h more than twice the speed of the other car. The angle between the roads is 120° . 3

After 3 hours, if the distance between the cars is 500 km, find the speed of the slower car, correct to one decimal place.

End of Question 12

Question 13 (13 marks) Use a SEPARATE writing booklet.

- (a) The graph of $y = f(x)$ is shown below. On a half-page diagram, sketch the graph of $y = f'(x)$, showing all important features. 2



- (b) The lines $l_1 : x - 2y - 5 = 0$ and $l_2 : 3x - y + 2 = 0$ intersect at P .

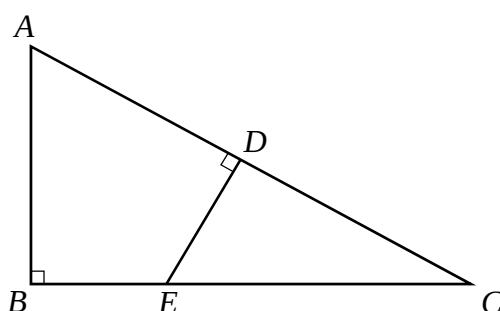
- (i) Show that the family of lines through P can be written in the form 1

$$x(1+3k) - y(2+k) + (2k-5) = 0$$

- (i) Hence find the equation of the line through P , parallel to the line $4x - y - 1 = 0$. 2

- (c) Find all solutions of $2\cos^2 x + 3\sin x = 3$, where $0^\circ \leq x \leq 360^\circ$. 3

- (d) $\triangle ABC$ is right-angled at B and DE is perpendicular to AC .



NOT TO SCALE

- (i) Prove that $\triangle ABC$ and $\triangle EDC$ are similar. 2

- (ii) Explain why $BC \times CE = AC \times CD$. 1

- (iii) Prove that $DE^2 = AD \times DC - BE \times EC$. 2

1 What is the value of $\sqrt{\frac{5\pi}{7}}$, correct to 2 significant figures?

(A) 1.5

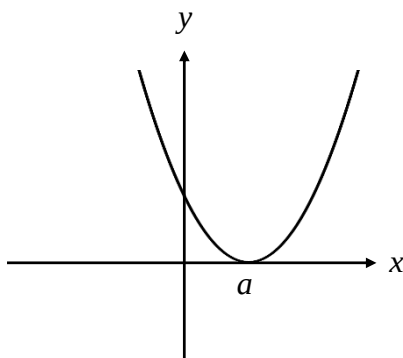
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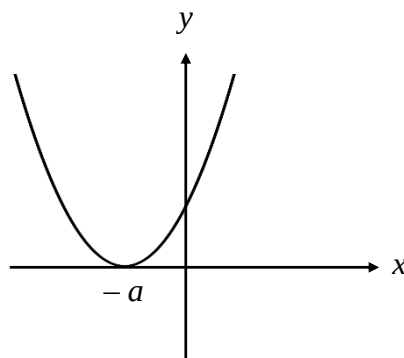
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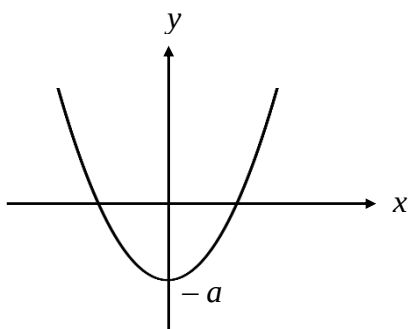
(A)



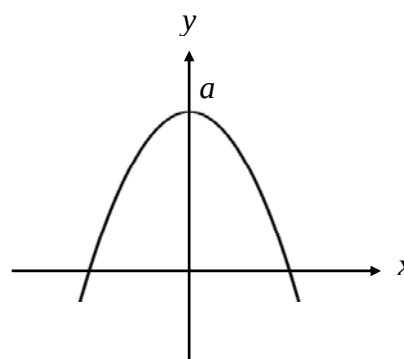
(B)



(C)



(D)



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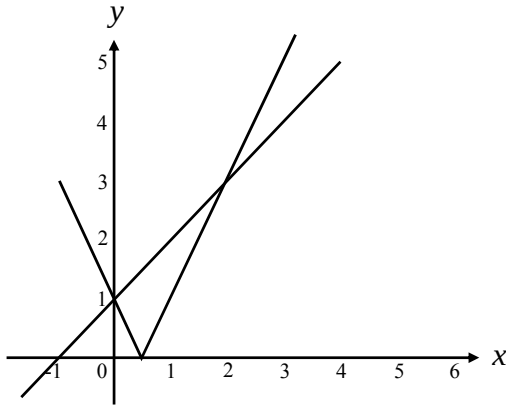
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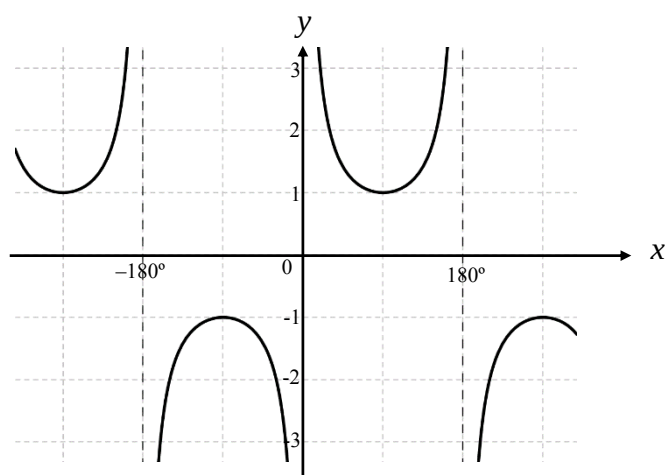
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(C) $y = \cot x$

(D) $y = -\sec x$

Question 8 (13 marks) Use a SEPARATE writing booklet.

- (a) Factorise $8m^3 - 1$. 2

$$(2m-1)(4m^2+2m+1)$$

- (b) Rationalise the denominator of $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}}$. 2

$$\begin{aligned} & \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} \\ &= \frac{(\sqrt{3})^2 - 2 \times \sqrt{3} \times \sqrt{2} + (\sqrt{2})^2}{(\sqrt{3})^2 - (\sqrt{2})^2} \\ &= 5 - 2\sqrt{6} \end{aligned}$$

- (c) Find the points of intersection of the line $x - y + 3 = 0$ and the hyperbola $xy = 10$. 3

$$x - y + 3 = 0 \quad \dots\dots (1)$$

$$xy = 10 \quad \dots\dots (2)$$

From (1),

$$y = x + 3 \quad \dots\dots (3)$$

Substitute (3) into (2)

$$x(x+3) = 10$$

$$x^2 + 3x - 10 = 0$$

$$(x+5)(x-2) = 0$$

$$\therefore x = -5 \text{ or } x = 2$$

$\therefore$ points of intersection are $(-5, -2)$ and $(2, 5)$

- (d) Find the centre and radius of the circle $x^2 - 6x + y^2 + 4y + 8 = 0$. 3

$$x^2 - 6x + y^2 + 4y + 8 = 0$$

$$(x^2 - 6x + 9) + (y^2 + 4y + 4) = -8 + 9 + 4$$

$$(x-3)^2 + (y+2)^2 = 5$$

$\therefore$ The radius is $\sqrt{5}$ and the centre is $(3, -2)$

(e) For what values of m does the equation $6x^2 - 12mx - 7m + 3 = 0$ have equal roots?

3

$$6x^2 - 12mx - 7m + 3$$

$$\begin{aligned}\Delta &= (-12m)^2 - 4(6)(-7m + 3) \\ &= 144m^2 + 168m - 72 = 0\end{aligned}$$

Equal roots occur when $\Delta = 0$

$$144m^2 + 168m - 72 = 0$$

$$6m^2 + 7m - 3 = 0$$

$$(3m - 1)(2m + 3) = 0$$

$$\therefore m = \frac{1}{3} \text{ or } m = -\frac{3}{2}$$

End of Question 8

Question 9 (13 marks) Use a SEPARATE writing booklet.

(a) Differentiate:

(i) $y = 2x^3 + 8x$ **1**

$$y = 2x^3 + 8x$$

$$\frac{dy}{dx} = 6x^2 + 8$$

(ii) $y = \frac{x^2}{1-x}$ **2**

$$y = \frac{x^2}{1-x}$$

$$\frac{dy}{dx} = \frac{(2x)(1-x) - (-1)(x^2)}{(1-x)^2}$$

$$= \frac{2x - 2x^2 + x^2}{(1-x)^2}$$

$$= \frac{2x - x^2}{(1-x)^2}$$

(iii) $y = 10x(3x-1)^5$ **3**

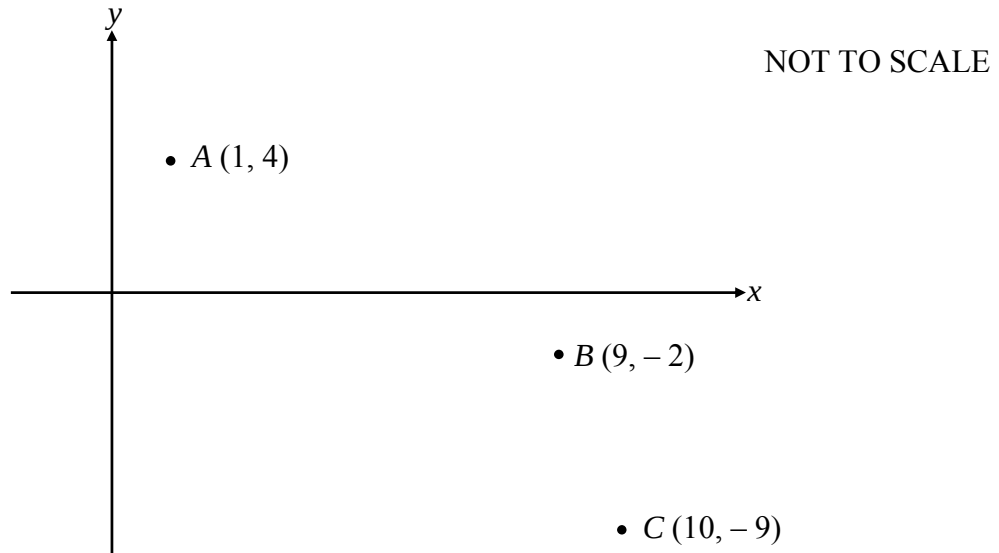
$$y = 10x(3x-1)^5$$

$$\frac{dy}{dx} = 10(3x-1)^5 + 10x \times 5 \times (3x-1)^4 \times 3$$

$$= 10(3x-1)^4((3x-1) + 15x)$$

$$= 10(3x-1)^4(18x-1)$$

- (b) It is given that points A , B and C have coordinates $(1, 4)$, $(9, -2)$ and $(10, -9)$ respectively.



- (i) Find the distance between A and B . 1

$$\begin{aligned}d_{AB} &= \sqrt{(1-9)^2 + (4-(-2))^2} \\&= \sqrt{64+36} \\&= 10 \text{ units}\end{aligned}$$

- (ii) Find the equation of the line that is parallel to AB and passes through C . 2

$$\begin{aligned}m_{AB} &= \frac{-2-4}{9-1} \\&= \frac{-6}{8} = -\frac{3}{4}\end{aligned}$$

$$\begin{aligned}y - (-9) &= -\frac{3}{4}(x - 10) \\4y + 36 &= -3x + 30 \\3x + 4y + 6 &= 0\end{aligned}$$

- (iii) Hence, show that the perpendicular distance between C and the line AB is 5 units. 1

$$\begin{aligned}\perp d &= \frac{|3(1) + (4)(4) + (6)|}{\sqrt{(3)^2 + (4)^2}} \\&= \frac{25}{5} = 5\end{aligned}$$

- (iv) Point D is placed so that $AB \parallel DC$. Given that quadrilateral $ABCD$ has area of 50 u^2 , show that the length of CD is 10 units.

2

$$50 = \frac{5}{2} \times (AB + CD)$$

$$10 + CD = 20$$

$$\therefore CD = 10$$

- (v) Find the coordinates of D .

1

$$AB = CD \text{ and } AB \parallel CD$$

$\therefore ABCD$ is a parallelogram since there is one pair of opposite sides that are equal and parallel

Diagonals AC and BD bisect each other.

$$\begin{aligned} M_{AC} &= \left(\frac{1+10}{2}, \frac{4+(-9)}{2} \right) \\ &= \left(\frac{11}{2}, -\frac{5}{2} \right) \end{aligned}$$

$$M_{BD} = \left(\frac{11}{2}, -\frac{5}{2} \right)$$

let D have coordinates (a, b)

$$\frac{9+a}{2} = \frac{11}{2} \text{ and } \frac{-2+b}{2} = -\frac{5}{2}$$

$$\therefore a = 2 \text{ and } b = -3$$

$$\therefore D \text{ has coordinates } (2, -3)$$

End of Question 9

Question 10 (13 marks) Use a SEPARATE writing booklet.

- (a) The gradient of the tangent to the curve $y = ax^3 + bx^2$ at $(1, 4)$ is 5.

3

Find the values of a and b .

$$y = ax^3 + bx^2$$

$$\frac{dy}{dx} = 3ax^2 + 2bx$$

$$\text{when } x = 1, \frac{dy}{dx} = 5$$

$$\text{also, when } x = 1, y = 4$$

$$5 = 3a + 2b$$

$$4 = a + b$$

solving simultaneously,

$$a = -3 \text{ and } b = 7$$

- (b) Find all solutions of $\sin 2x = \frac{1}{2}$, where $0^\circ \leq x \leq 360^\circ$.

3

$$\sin 2x = \frac{1}{2}$$

$$2x = 30^\circ, 150^\circ, 390^\circ, 510^\circ$$

$$x = 15^\circ, 75^\circ, 195^\circ, 255^\circ$$

- (c) Find values of a , b and c , for which $-x^2 + 4x - 3 \equiv a(x+3)^2 + b(x+3) + c$.

2

$$-x^2 + 4x - 3 \equiv a(x+3)^2 + b(x+3) + c$$

$$\text{RHS} = a(x+3)^2 + b(x+3) + c$$

$$= a(x^2 + 6x + 9) + bx + 3b + c$$

$$= ax^2 + 6ax + 9a + bx + 3b + c$$

$$= ax^2 + (6a + b)x + (9a + 3b + c)$$

Equating coefficients,

$$a = -1$$

$$6a + b = 4$$

$$\therefore b = 10$$

$$9a + 3b + c = -3$$

$$\therefore c = -24$$

$$\therefore -x^2 + 4x - 3 \equiv -(x+3)^2 + 10(x+3) - 24$$

Alternatively,

Equate coefficients, $a = -1$

Substitute $x = -3$,

$$-9 - 12 - 3 = c$$

$$\therefore c = -24$$

Substitute $x = 0$,

$$-3 = -9 + 3b - 24$$

$$3b = 30$$

$$\therefore b = 10$$

$$a = -1, b = 10 \text{ and } c = -24$$

(d) Solve for x : $2^x < 5$.

2

$$2^x < 5$$

$$\log(2^x) < \log 5$$

$$x \times \log 2 < \log 5$$

$$x < \frac{\log 5}{\log 2}$$

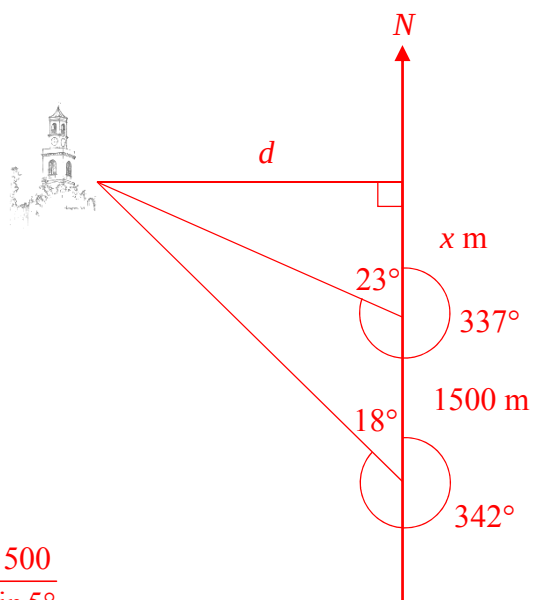
$$x < 2.321928095\dots$$

$$x < 2.32 \text{ (2 decimal places)}$$

(e) A man walking due north along a level road observes a church spire on a bearing of 342° T. After walking 1500 metres, he observes on a bearing of 337° T.

3

Find the perpendicular distance from the church to the road.



$$\frac{y}{\sin 18^\circ} = \frac{1500}{\sin 5^\circ}$$

$$y = \frac{1500 \sin 18^\circ}{\sin 5^\circ}$$

$$\sin 23^\circ = \frac{d}{y}$$

$$d = \frac{\sin 23^\circ \times 1500 \sin 18^\circ}{\sin 5^\circ}$$

$$= 2078 \text{ m (nearest metre)}$$

End of Question 10**Question 11** (13 marks) Use a SEPARATE writing booklet.

(a) If α and β are the roots of the equation $2x^2 - 3x + 5 = 0$, find the value of the following:

(i) $\alpha + \beta$ 1

$$\alpha + \beta = \frac{3}{2}$$

(ii) $\alpha\beta$ 1

$$\alpha\beta = \frac{5}{2}$$

(i) $(\alpha + 2)(\beta + 2)$ 2

$$\begin{aligned}(\alpha + 2)(\beta + 2) &= \alpha\beta + 2(\alpha + \beta) + 4 \\&= \frac{5}{2} + 2\left(\frac{3}{2}\right) + 4 \\&= \frac{19}{2}\end{aligned}$$

(b) Prove that $(\sec \theta - \cos \theta)^2 = \tan^2 \theta - \sin^2 \theta$. 3

$$\begin{aligned}\text{LHS} &= (\sec \theta - \cos \theta)^2 \\&= \sec^2 \theta - 2 \sec \theta \cos \theta + \cos^2 \theta \\&= (1 + \tan^2 \theta) - \frac{2 \cos \theta}{\cos \theta} + (1 - \sin^2 \theta) \\&= 1 + \tan^2 \theta - 2 + 1 - \sin^2 \theta \\&= \tan^2 \theta - \sin^2 \theta = \text{RHS}\end{aligned}$$

- (c) (i) Derive the equation of the locus of all points $P(x, y)$ which are equidistant from the origin O and the line $y = 4$. 2

$$P(x, y) \quad l: (x, 4) \quad O(0, 0)$$

$$\begin{aligned} d_{Pl} &= \sqrt{(x-x)^2 + (y-4)^2} \\ &= y-4 \end{aligned}$$

$$\begin{aligned} d_{PO} &= \sqrt{(x-0)^2 + (y-0)^2} \\ &= \sqrt{x^2 + y^2} \end{aligned}$$

$$d_{Pl} = d_{PO}$$

$$y-4 = \sqrt{x^2 + y^2}$$

$$y^2 - 8y + 16 = x^2 + y^2$$

$$x^2 = -8y + 16$$

$$x^2 = -8(y-2)$$

- (ii) Describe the locus geometrically. 1

Concave down parabola with focal length 2, and vertex $(0, 2)$.

- (d) Find the values of k for which the expression $kx^2 + (k-1)x + k$ is positive definite. 3

$$\Delta = b^2 - 4ac$$

$$= (k-1)^2 - 4(k)(k)$$

$$= k^2 - 2k + 1 - 4k^2 = -3k^2 - 2k + 1$$

$$\Delta < 0,$$

$$-3k^2 - 2k + 1 < 0$$

$$(-3k+1)(k+1) < 0$$

$$k < -1 \text{ or } k > \frac{1}{3}$$

$k > 0$, since the quadratic is positive definite

$$\therefore k > \frac{1}{3}$$

End of Question 11

Question 12 (13 marks) Use a SEPARATE writing booklet.

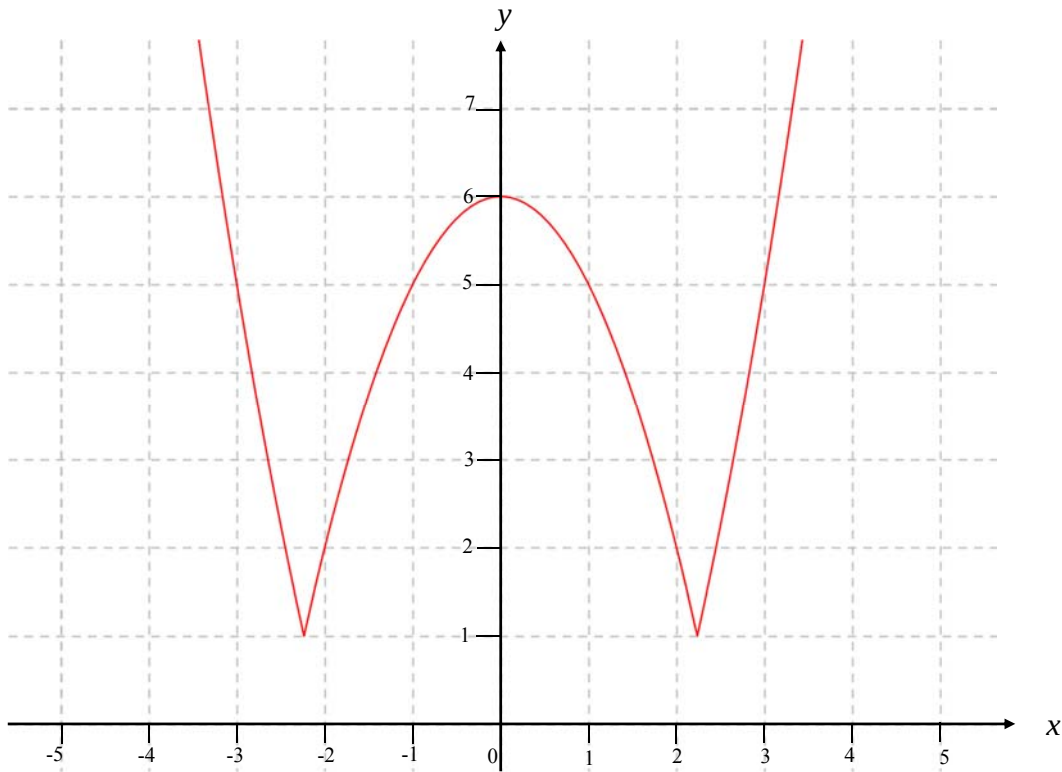
(a) Evaluate $\lim_{x \rightarrow 5} \frac{6x^2 - 29x - 5}{x - 5}$

2

$$\begin{aligned}\lim_{x \rightarrow 5} \frac{6x^2 - 29x - 5}{x - 5} &= \lim_{x \rightarrow 5} \frac{(x - 5)(6x + 1)}{x - 5} \\ &= \lim_{x \rightarrow 5} 6x + 1 \\ &= 31\end{aligned}$$

(b) Sketch $y = |x^2 - 5| + 1$, showing all important features.

2



(c) Consider the function $f(x) = x^2 - 5x + 2$.

(i) Differentiate $f(x)$ from first principles.

2

$$f(x) = x^2 - 5x + 2$$

$$\begin{aligned}f(x+h) &= (x+h)^2 - 5(x+h) + 2 \\ &= x^2 + 2xh + h^2 - 5x - 5h + 2\end{aligned}$$

$$\begin{aligned}f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 5x - 5h + 2 - x^2 + 5x - 2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(2x - 5 + h)}{h} \\ &= \lim_{h \rightarrow 0} 2x - 5 + h \\ &= 2x - 5\end{aligned}$$

- (ii) Find the equation of the normal to the curve $y = f(x)$ at the point where $x = 1$. 2

$$\text{At } x = 1, m_T = 2(1) - 5 = -3$$

$$\therefore m_N = \frac{1}{3}$$

$$\text{At } x = 1, y = 1^2 - 5(1) + 2 = -2$$

$$y - (-2) = \frac{1}{3}(x - 1)$$

$$x - 3y - 7 = 0$$

- (d) A parabola whose axis is parallel to the x -axis has vertex $(2, 3)$ and passes through $(-1, 9)$. 2

Find the equation of the parabola.

$$(y - 3)^2 = 4a(x - 2)$$

$$\text{let } x = -1 \text{ and } y = 9$$

$$36 = -12a$$

$$\therefore a = -3$$

$$\therefore (y - 3)^2 = -12(x - 2)$$

- (e) Two cars leave a point P at the same time. They travel away from each other on different straight roads. One car travels at 10 km/h more than twice the speed of the other car. The angle between the roads is 120° . 3

After 3 hours, if the distance between the cars is 500 km, find the speed of the slower car, correct to one decimal place.

$$\cos 120^\circ = \frac{(3x)^2 + 3^2(2x + 10)^2 - 500^2}{2 \times 3x \times 3(2x + 10)}$$

$$-\frac{1}{2} = \frac{9x^2 + 9(4x^2 + 40x + 100) - 250000}{18x(2x + 10)}$$

$$-\frac{1}{2} = \frac{9x^2 + 36x^2 + 360x + 900 - 250000}{18x(2x + 10)}$$

$$-9x(2x + 10) = 45x^2 + 360x - 249100$$

$$-18x^2 - 90x = 45x^2 + 360x - 249100$$

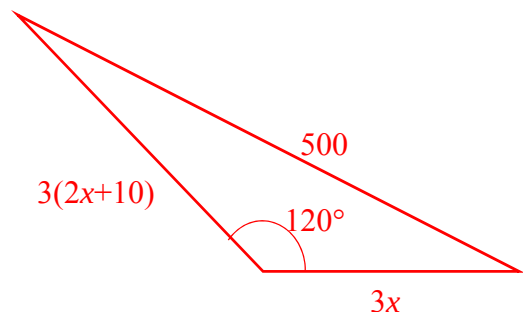
$$63x^2 + 450x - 249100 = 0$$

$$x = \frac{-450 \pm \sqrt{450^2 - (4)(63)(-249100)}}{2 \times 63}$$

$$\text{since } x > 0,$$

$$x = 59.410\dots$$

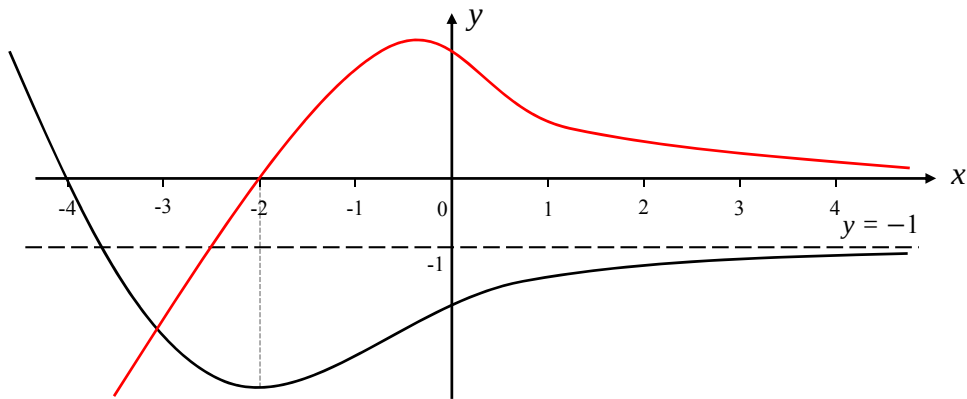
$$= 59.4 \text{ km (1 decimal place)}$$



End of Question 12

Question 13 (13 marks) Use a SEPARATE writing booklet.

- (a) The graph of $y = f(x)$ is shown below. On a half-page diagram, sketch the graph $y = f'(x)$, showing all important features. 2



- (b) The lines $l_1 : x - 2y - 5 = 0$ and $l_2 : 3x - y + 2 = 0$ intersect at P .

- (i) Show that the family of lines through P can be written in the form 1

$$x(1 + 3k) - y(2 + k) + (2k - 5) = 0$$

$$x - 2y - 5 + k(3x - y + 2) = 0$$

$$x - 2y - 5 + 3kx - ky + 2k = 0$$

$$x + 3kx - 2y - ky - 5 + 2k = 0$$

$$x(1 + 3k) - y(2 + k) + (2k - 5) = 0 \text{ as required}$$

- (ii) Hence find the equation of the line l through P , parallel to the line $4x - y - 1 = 0$. 2

the line parallel to $4x - y - 1 = 0$ has gradient 4

the gradient of $x(1 + 3k) - y(2 + k) + (2k - 5) = 0$ is $\frac{1 + 3k}{2 + k}$

$$\frac{1 + 3k}{2 + k} = 4$$

$$1 + 3k = 8 + 4k$$

$$k = -7$$

$$\therefore x(1 - 21) - y(2 - 7) + (-14 - 5) = 0$$

$$-20x + 5y - 19 = 0$$

- (c) Find all solutions of $2\cos^2 x + 3\sin x = 3$, where $0 \leq x \leq 360^\circ$.

3

$$2(1 - \sin^2 x) + 3\sin x = 3$$

$$2 - 2\sin^2 x + 3\sin x - 3 = 0$$

$$2\sin^2 x - 3\sin x + 1 = 0$$

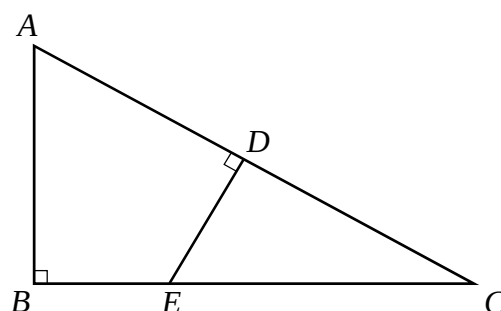
$$(2\sin x - 1)(\sin x - 1) = 0$$

$$\therefore \sin x = \frac{1}{2} \text{ or } \sin x = 1$$

in the domain $0^\circ \leq x \leq 360^\circ$

$$x = 30^\circ, 150^\circ \text{ or } 90^\circ$$

- (d) $\triangle ABC$ is right-angled at B and DE is perpendicular to AC .



NOT TO SCALE

- (i) Prove that $\triangle ABC$ and $\triangle EDC$ are similar.

2

In $\triangle EDC$ and $\triangle ABC$,

$\angle BCA$ is common

$\angle ABC = 90^\circ$ ($\triangle ABC$ is a right-angled triangle)

$\angle EDC = 90^\circ$ ($DE \perp AC$)

$\angle ABC = \angle EDC = 90^\circ$

$\therefore \triangle EDC \parallel \triangle ABC$ (equiangular)

- (ii) Explain why $BC \times CE = AC \times CD$.

1

$$\frac{CE}{AC} = \frac{CD}{BC} \text{ (corresponding sides of similar triangles)}$$

$$\therefore BC \times CE = AC \times CD$$

(iii) Prove that $DE^2 = AD \times DC - BE \times EC$.

2

$$AC = AD + DC$$

$$BC = BE + EC$$

from (i),

$$BC \times CE = AC \times CD$$

$$CE(BE + EC) = CD(AD + DC)$$

$$CE \times BE + CE^2 = AD \times CD + DC^2$$

$$EC^2 - DC^2 = AD \times DC - BE \times EC$$

by Pythagoras theorem,

$$DE^2 = EC^2 - DC^2$$

$$\therefore DE^2 = AD \times DC - BE \times EC \text{ as required}$$

End of Paper